

Photon-Ring Retarded-Time Tomography: The Mahakal Phenomenon

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Prepared under ruling `PAPER_I_FINAL_TEXT_REVIEW_039` , which accepted the 038 method corrections, resolved two archived method dependencies from source, and authorized a bounded publication-preparation pass. The archived Shiva-era manuscript at `artifacts/manuscript/PAPER_I.md` , the working revisions 036, 037 and 038, the canonical freeze and every run directory are unchanged. Every number below is traced in `CLAIM_EVIDENCE_MATRIX_039.json` to a primary artifact, a commit, a hash and a row selector. One execution input — the realized value of the R1 metric floor — remains unlocated and is carried as a named limitation rather than filled in; see section 9 and `METHOD_DEPENDENCY_STATUS_039.json` .

Preprint draft — theory and controlled synthetic computation only. No telescope detection, no laboratory result, and no recovery from a resolved real photon ring is claimed anywhere.

Abstract

Near-critical black-hole light paths sample a variable source at distributed retarded times. We distinguish **historical reach**, the number of **source directions supported** by a declared observation model, and **held-out recovery** of a specified historical object. We call their separation the Mahakal phenomenon.

For a separable source representation, a temporal factor whose support is disjoint from the direct retarded-time footprint generates exactly zero direct columns; higher-order samples can make such columns nonzero, but those nonzero columns need not be independent or operationally recoverable.

In the archived 12-geometry finite-model audit, ideal order-labelled observations extend the anchor-connected detectable interval beyond direct imaging. At the reference geometry the sampled direct operator annihilates a 72-dimensional old-contrast target, while the order-labelled stack retains two operational combinations after profiling the other 152 coefficients of the full L224 model. That count uses the frozen legacy measure; the available row-reweighting bound permits one to two combinations under its stated assumptions.

Two sealed reconstruction benchmarks address different objects. Baseline-inclusive emissivity-level span increases from 48 to 80 M in the reported in-class regimes and the mild-mismatch diagnostic, while severe mismatch remains 0 to 0 M. A separate 60-history morphology benchmark reports paired gains of 0.133 for TSVD and 0.164 for ridge in its registered error-reduction metric, with lower confidence bounds 0.101 and 0.116. These aggregate gains do not establish reliable multiple-feature recovery or a nonzero stable morphology interval.

The numerical results are conditional on the specified finite operators, source spaces, noise conventions and evaluation metrics. Continuous transfer quadrature remains unqualified, and the archived index-sum control does not establish a physical order-resolution advantage. The study delineates a bounded information and recovery problem rather than demonstrating reconstruction of a physical movie.

1. Introduction

A photon emitted near the equatorial plane of a Kerr black hole can reach a distant observer directly or after one or more near-critical windings. Each additional half-orbit adds delay, so the n -th image shows the source as it was further in the past.

Three questions are easy to conflate, and this paper keeps them apart:

1. **Historical reach** (section 5) — which earlier source times are sampled at all, and which of those are detectable at a declared SNR.
2. **Historical dimension** (section 6) — how many directions of a declared finite source model the observation supports, before and after profiling nuisance coefficients.
3. **Historical recovery** (section 7) — whether a held-out truth can be reconstructed to a preregistered materiality floor.

1.1 The Mahakal phenomenon

We use the name for the separation itself. Higher-order paths can sample source-time regions the direct image misses. Increasing source resolution can expose null or weak directions a coarser model cannot represent. **Neither statement says that source enrichment always destroys injectivity, or that compact temporal support alone makes a mode invisible.** The support result of section 4 is conditional on disjointness from the direct footprint, and the rank behaviour of section 6 is a finding about the tested models.

1.2 Relation to prior work

The demagnified, rotated and delayed higher-order image hierarchy is established by Kerr-lensing studies [GL2020], and its interferometric signatures have been analyzed independently [J2020]. AART provides the ray-tracing framework underlying the archived maps [AART2023]. Photon-ring correlation methods [H2021], hotspot-based spacetime tomography [T2020], dynamic VLBI reconstruction [B2018], and orbital polarimetric flare tomography [L2024] address related temporal or inverse problems. Here we isolate, in a declared finite source representation and fixed observation model, the distinction between source-time sensitivity, nuisance-adjusted supported directions, and held-out recovery of specified source objects.

That last sentence describes this paper's scope. It is not a priority claim. Section 11 states exactly which parts of each cited record were read; no external paper validates the computations reported here, and none is described as fully read where only its abstract and metadata were verified.

1.3 What is not claimed

- No detection; every number is synthetic.
- No recovery of a historical movie.
- No qualified continuum accuracy for the transfer integral (section 8).
- No physical order-resolution advantage. The `UNRESOLVED_IMAGE` arm is a named **compression control**, not a physically unresolved image.
- No claim that conditioning is evidence of physical content: a pairing-permuted control is better conditioned than the physical operator at 12 of 12 geometries across 16 frozen seeds.

2. Forward model

For a fixed geometry, the order- n transfer model maps emissivity `j` to an order-specific field on the screen,

$$(T_n j)(\xi, t_o) = \chi_n(\xi) \cdot w_n(\xi) \cdot j(r_n(\xi), \phi_n(\xi), t_o - \Delta_n(\xi))$$

where `χn` is the order's domain mask, `wn` the ray weight, and `Δn` the retarded delay. The ray-intensity factor contains `g3` under the declared model; geometric area and measurement noise are specified separately.

Time convention. Within the D026 acquisition audit of section 8, delays are referred to one absolute clock reference $T_{REF} = -978.6055123201214$ M; three archived profiles had each recentred their own reference, differing by up to 0.69 M, and that audit reconciles them. This is a convention **imposed in that audit**. It does not mean every legacy result was retrospectively recentred, and the earlier experiments retain the clock they were run under.

Measurement convention. The whitened row carries $\sqrt{d\Omega} \cdot g^3$. An earlier flat-per-row convention made Fisher information scale with pixel count — a discretisation artefact — and is retired. Results computed under it are recomputed or labelled legacy in place, never silently reinterpreted.

Acquisition unit. For detector cell D_d and **screen integration cell** C_p intersected with the order's effective domain Ω_n — both live in the same screen coordinate space; the source-plane map appears inside the transfer function and is not a set intersected directly with the observer's pixel —

$$O[d,p] = |D_d \cap C_p \cap \Omega_n|, \text{ an area}$$
$$C_{inherited} = \sigma^2 \cdot O \cdot \text{diag}(1/a) \cdot O^T, a_p = |P_p|$$

so brightness maps to integrated flux. The whole-cell emitting fraction times the uncut overlap is **not** the clipped overlap: a unit cell emitting on its leftmost fifth and straddling two half-width pixels owes (0.2, 0) and that substitution pays (0.1, 0.1).

3. Source spaces, observation model, metrics and estimators

Each experiment pins its own. They are not interchangeable, and no single noise convention covers all of them.

experiment	source space	evaluation	noise convention
E3C geometry audit	declared temporal/spatial product class on the common age grid	age-threshold detectability masks	white noise of density σ_Ω per unit solid angle, $\text{Var}(\eta_p) = \sigma_\Omega^2 d\Omega_p$, one density for the whole audit; per-geometry SNR sweep
R2 reference geometry	full L224, 72-dim old- contrast target, 152-dim nuisance	normalised Fisher traces and operational counts	frozen legacy measure, $\sigma =$ 0.011341986814407566
R1 sealed main	C224	baseline-inclusive field metric, anchored stable span	one resolved Gaussian draw per (truth, SNR, draw index) at $\text{SNR}_0 = 100$; the direct, unresolved and total-flux observations are declared linear readouts of that same draw
R1L stage- 2R	localized L224/L448/L1056	aggregate structural materiality	as R1, at $\text{SNR}_0 = 1000$
HMT-2 sealed main	L448_contrast, L896_radial_enriched	registered morphology error-reduction, paired draws	one standard-normal tensor of shape (orders, rays, observer times) per truth and draw, mapped to σ_Ω $\sqrt{d\Omega} z$ on each order's pixels and then through each arm's declared readout, whitened by the channel variance
D026 acquisition audit	detector-space response	whitened detector-vector relative error	single-sky σ^2 over full pixel area once

R2's ideal order-labelled comparison is not the later single-sky detector experiment, and the two must not be described under one noise pin. Replication is a separate axis from the noise law: R1 draws eight noisy realisations per truth plus one noiseless control that the endpoint excludes, and HMT-2 draws four paired realisations per truth. A draw count is not an observation covariance.

3.1 The declared source class and observation schedule

C224 is $4 \times 7 \times 8 = 224$: four cubic B-splines in $\log r$, seven real Fourier modes in azimuth ($|m| \leq 3$), and eight DCT modes in source time, ordered radial-major then azimuthal then temporal. Its declared support is $r \in [1.8660386527060988, 49.98205255591607] \text{ M}$ and $t \in [-128.82234649196255, 29.0] \text{ M}$. The reference observer schedule is eight samples uniformly spaced from 0 to 20 M, with 1536 retained rays per order at orders $n = 0, 1, 2$.

The localized L-classes replace the DCT temporal factor by a compactly supported temporal representation; they are a different temporal representation, not a renaming of C224, and each carries its own definition. L896_radial_enriched is $8 \times 7 \times 16$ and L448_contrast is $4 \times 7 \times 16$; the HMT-2 runner removes $m = 0$ before inversion, leaving 768 and 384 fitted contrast coefficients.

3.2 The age probe and the detectability convention

The E3C localized probe is Gaussian in retarded age and flat in the emission annulus, with half width $h = 3.0 \text{ M}$, **normalised to unit L^2 norm over the emission region** so that the reported Fisher information is not in units of an arbitrary peak amplitude. The common age grid has step 4 M and maximum 252 M , fixed source-independently before any geometry was evaluated.

Detectability is a threshold on the operator scaled to the reference SNR:

$$\text{oldest detectable age probe} = \sup \{ a : \text{SNR}_0^2 \cdot I(a) \geq \rho^2 \}, \rho = 1$$

with $I(a)$ the Fisher information for the unit-norm probe at age a . SNR_0 is a scaling of the whitened operator; the noise density σ_Ω is fixed once, from the direct arm's clean response to the declared reference source, and no arm chooses its own σ . The reference value is $\text{SNR}_0 = 100$, and the frozen sweep runs 1 to 10^6 in half-decades. Reporting a reference SNR alone would not define the experiment; the definition above is what the retained E3C quantities use.

3.3 Estimators and where their tuning came from

TSVD and ridge are non-Bayesian spectral estimators with a fixed representation and validation-selected regularization. **The tuning lineage is per experiment, not global.** R1 reuses the hyperparameters selected on the R0C repair-validation bank and frozen before any main-test truth. HMT-2 reuses its own stage-1 selection, inherited unchanged through the HMT-2 sealed-main freeze, whose runner performs no sweep. Neither main experiment tunes on its held-out bank. Each result names which estimator produced it. (The earlier word "prior-free" is replaced; no estimator and no selected value changed.)

Three rank notions, kept distinct, and recovery is not one of them. *Structural algebraic rank* where it is exactly justified; *numerical rank* at a stated floating-point tolerance; *operational rank* at the declared SNR, source normalisation and decision threshold. **Estimator recovery** is a separate criterion — a declared loss and a reliability requirement on held-out histories. A nonzero column is none of the four.

Nuisance projection is part of the stated R2 target calculation in section 6. It is not an automatic step applied to the earlier E3C and E3D operational counts, which are taken at their own thresholds without a target/nuisance split.

4. The support result

Let sampled source functions be separable, $q_{lk}(r, \phi, t) = \psi_l(r, \phi) \tau_k(t)$, and let W_n be the set of retarded times reached by the actual observer times and retained rays of order n with nonzero transfer weight.

Proposition 4.1 (sampled support-nullity). If $\text{supp}(\tau_k) \cap W_0 = \emptyset$ then $A_0 q_{lk} = 0$ for every spatial factor l . If a retained higher-order row has nonzero weight, nonzero spatial value and nonzero temporal value, the corresponding stacked column is nonzero. If m such temporal functions and d_s spatial factors are independent, they supply $m d_s$ linearly independent coefficient directions in the direct null space.

Proof. Each row of A_0 is the product of the ray weight, the spatial value at the ray's landing point and the temporal value at the ray's retarded time. Disjoint temporal support makes the last factor zero at every direct sample, so every such row vanishes and the

column is identically zero. Conversely a single retained higher-order row with all three factors nonzero has a nonzero product, so that stacked column is nonzero. Independence of the selected basis products gives the nullity count. ■

What the proposition does not say. A nonzero column need not be an independent output direction, and need not be an operationally recoverable one. The proof establishes nonvanishing, not sufficiency: no independence among the newly nonzero columns follows from it. The theorem and its nonvanishing assumptions are unchanged.

Corollary 4.2 (conditional continuum statement). Under the assumed transfer map, a temporal factor whose support is disjoint from the full supported screen-time footprint — not its finite sampled proxy — produces zero response almost everywhere. A nonzero continuum response requires non-vanishing on a set of positive measure, not an isolated support intersection.

Corollary 4.2 is a statement about the assumed map. It is **not** a validation of that map's discretisation; see section 8.

5. Historical reach

5.1 Reach is three statistics, not one

Following the age-interval semantics amendment, we report:

- **oldest detectable age probe** — the supremum of the detectable age set. A supremum cannot distinguish history detectable all the way back from a detectable island beyond a gap.
- **longest detectable run** — the longest run of consecutive detectable ages anywhere on the grid; not a depth from the present.
- **anchor-connected span** — the stretch that reaches the frozen anchor: a contiguous passing interval beginning at the geometry-specific anchor. It begins at age zero only where the recorded anchor is zero, which is 8 of the 12 geometries; at the four 75-degree geometries the anchor is 32, 28, 28 and 28 M and the interval begins there.

Each geometry is anchored at its own youngest fully supported probe centre, fixed before any detectability curve was read, because there the minimum delay in the frozen ray set exceeds the last observer sample.

5.2 The archived reach statistics

Oldest detectable age probe (M), resolved stack, then direct: 60, 84, 144 at $i = 20, 50, 75$ for all four spins; 20, 60, 140 for the direct image.

Anchor-connected span (M), resolved stack: 60 and 84 at 20 and 50 degrees, and at 75 degrees 112 M at spin 0 and 116 M at spins 0.50, 0.90 and 0.98. **Direct:** 20 and 60, and at 75 degrees 108 M at spin 0 and 112 M at the other three.

At the reference SNR the resolved stack's oldest detectable probe exceeds the direct image's at 12 of 12 geometries, and the threshold-independent innovation is positive at 12 of 12. Figure 1 plots the two statistics side by side.

Two reach statistics, and only one of them is flat in spin

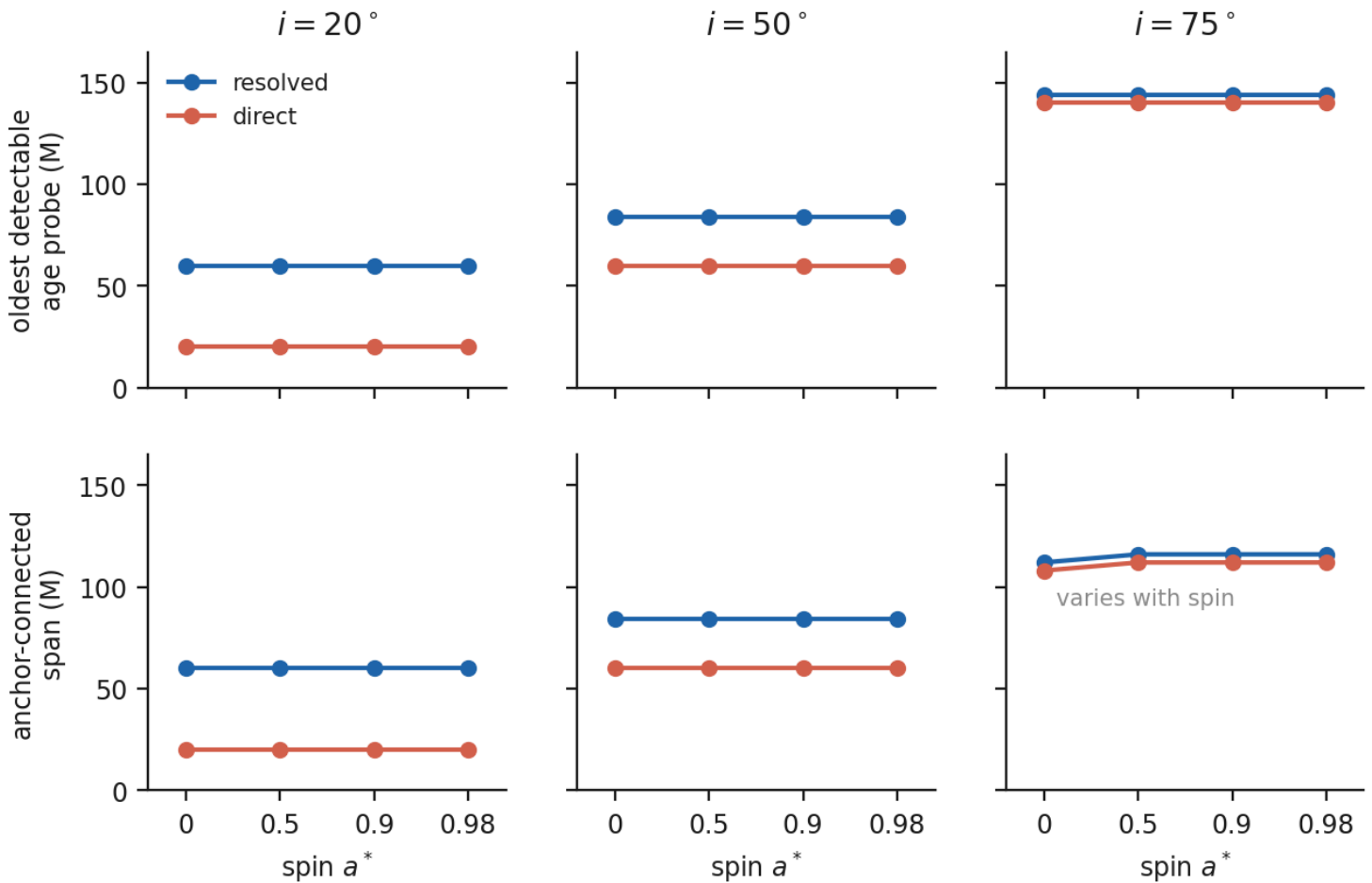


Figure 1. Two reach statistics, and only one of them is flat in spin. Top row: the oldest detectable age probe, the supremum of the detectable age set, which is flat across the four sampled spins at every inclination. Bottom row: the anchor-connected span, the contiguous passing interval from each geometry's own anchor, which is not — at 75 degrees the resolved span is 112 M at spin 0 and 116 M at the other three. A supremum being flat in spin does not establish spin independence of the reach that connects to the anchor. Reference SNR, median over source class; regenerated from `artifacts/tables/e3c_depth_curves.parquet` with the selector recorded in `FIGURE_MANIFEST_039.json`.

Scope. 144 M is a supremum of a threshold mask. It is not an estimator recovery depth and not an anchored interval. **A common oldest passing grid point across four spins does not establish physical spin independence** — the anchor-connected quantities themselves vary with spin at 75 degrees.

5.3 Delay and spatial substitutions

The archived 0.98 and 0.57 comparisons transplant independently sampled arrays by index. We retain them as **index-paired counterfactuals**. They do not isolate the physical contributions of co-registered delay and spatial remapping, and no mechanism claim is made from them.

6. Historical dimension

6.1 Definitions

Write $y = A_T c_T + A_N c_N + \varepsilon$ with the declared positive-definite noise model C , and set $B_T = C^{(-1/2)} A_T$, $B_N = C^{(-1/2)} A_N$. Let $H_T = R^T R$ be the target source Gram and Π_N the orthogonal projector onto $\text{range}(B_N)$. In source-normalised target coordinates,

$$\begin{aligned} B_{\text{known}} &= B_T R^{(-1)} \\ B_{\text{cond}} &= (I - \Pi_N) B_T R^{(-1)} \\ F_{\text{known}} &= B_{\text{known}}^T B_{\text{known}}, F_{\text{cond}} = B_{\text{cond}}^T B_{\text{cond}} \end{aligned}$$

The quoted information totals are traces of these normalised Fisher matrices. Operational directions are counted from the singular values at the declared amplitude threshold. **If σ is already inside C , the singular values are not multiplied by the SNR label a second time.** This is a target/nuisance information calculation, not a reconstruction estimator.

6.2 The reference geometry

At `a050_i050`, target dimension 72, $\sigma = 0.011341986814407566$. The nuisance space has **152 columns**; its sampled rank is **140 in the direct arm** and **152 in the stacked resolved arm**. Column count and arm-specific rank are different numbers and are labelled separately:

arm	tr F known	tr F conditional	operational at $\rho = 1$
DIRECT_PHYSICAL	0	0	0
RESOLVED_PHYSICAL	11.053660	7.198318	3 $\rightarrow$ 2

Singular values 1.4371649, 1.3458733, 0.8062804; the third misses $\rho = 1$ by a clear margin, and subspace angles are exactly zero across all three rank tolerances, so neither the count nor the subspace is a tolerance artefact.

Measure pin. The count 2 belongs to the **frozen legacy measure**. The available row-reweighting bound permits **one to two** combinations under its stated row-noise and metric assumptions, because the per-order totals do not bound the individual rows: the order-0 row ratios span `[0.500081, 1.000162]`. `tr F_cond / tr F_known = 0.6512158` is a ratio of normalised Fisher traces in the declared source norm — not bits, not a fraction of a history.

6.3 Where the two combinations live

The two operational combinations are **temporally broad rather than confined to one epoch**. Using their exported source coefficients and retaining the cross terms of the source Gram, their squared source norms place

source-time interval (M)	first mode	second mode
[−128.82, −109.09] oldest third	2.44%	2.35%
[−109.09, −89.37] middle third	43.81%	43.92%
[−89.37, −69.64] youngest third	53.75%	53.73%

About 97.6% lies in the younger two thirds, **but the oldest-third contribution is not zero**. These localizations describe signed source perturbations, not separately recovered frames.

6.4 Rank behaviour under enrichment

The E3D ladder is four classes, and it changes **both** spatial and temporal factors — it is not temporal enrichment alone:

class	radial × azimuthal × temporal	dimension	median operational rank	as a fraction	σ_min+
C224	4 × 7 × 8	224	201	0.897	1.635e-02
C448_T	4 × 7 × 16	448	365	0.815	7.164e-05
C528_S	6 × 11 × 8	528	441	0.835	2.871e-04
C1056_ST	6 × 11 × 16	1056	770	0.729	1.332e-07

Over the full ladder C224 to C1056_ST the median operational-rank **fraction** falls 0.897 to 0.729 — the 16.8 percentage points — while the operational rank itself **rises**, 201 to 770. A falling supported fraction is compatible with an increasing absolute rank, and the two must not be quoted as one statement. σ_min+ falls about five orders of magnitude across the same ladder. The resolved operator's *numerical*-rank fraction falls by up to 3.3%, with the largest fall over any arm 25.1% at DIRECT_PHYSICAL , a000_i020 — a different quantity again. Figure 2 plots the fraction, the count and the localized-directional depth across the same four classes.

The supported fraction falls while the count rises

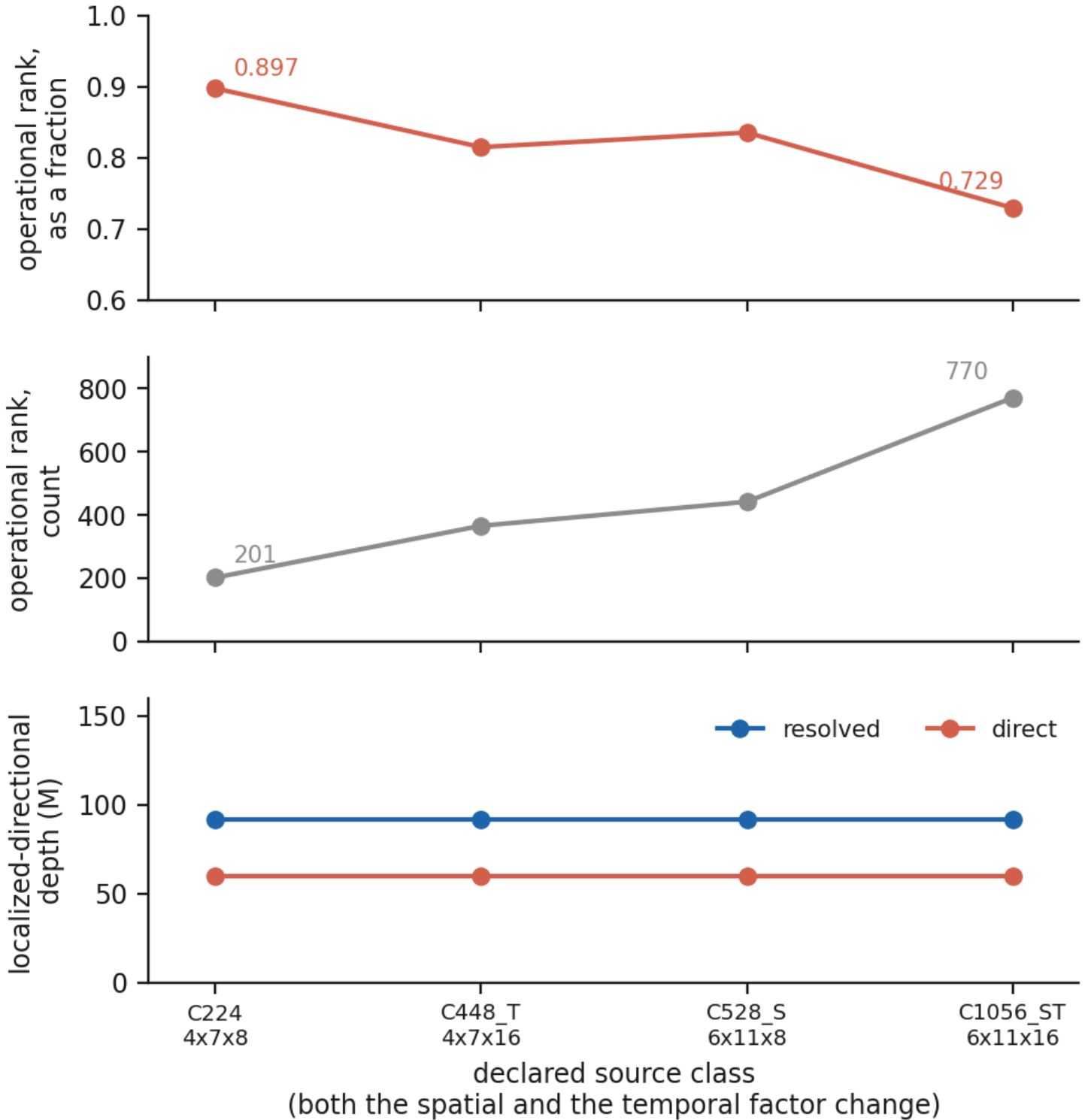


Figure 2. The supported fraction falls while the count rises. Enrichment across the four declared source classes; tick labels give radial $\times$ azimuthal $\times$ temporal, so both factors are visibly changing along the ladder. Top: the median operational rank as a fraction of class dimension falls 0.897 to 0.729. Middle: the operational rank itself rises, 201 to 770. Bottom: the localized-directional depth, the deepest retarded age whose best-determined localized mode clears the operational threshold at the reference SNR, which is flat across the ladder for both physical arms; across all physical and mechanism arms the largest move anywhere in this table is a single 4 M grid step. This is not the scalar detectability probe of figure 1, and the two

must not be identified because both are quoted in M. Regenerated from `artifacts/tables/e3d_class_spectra.parquet` and `artifacts/tables/e3d_depth_by_class.parquet` ; selectors in `FIGURE_MANIFEST_039.json` .

Over the same ladder the localized-directional depth moves by at most one grid step of 4 M among the physical and mechanism arms. This is not the scalar E3C detectability probe of section 5.2, and the two must not be identified because both are quoted in M.

The asymmetry is worth stating in its own right: every move in that depth table appears at the **spatial** enrichment steps (`C224` to `C528_S` , `C448_T` to `C1056_ST`) and none at the temporal one. Temporal enrichment exposes null directions without extending reach; spatial enrichment extends reach without exposing many.

These are findings for the tested models, not a theorem that enrichment destroys injectivity.

7. Historical recovery

7.1 Level: the R1 sealed main

The bank holds 640 histories across four regimes, committed by hash before scoring. At `SNR_0 = 100` , with both TSVD and ridge, the anchored baseline-inclusive level span moves:

regime	direct (M)	resolved (M)	gain (M)	material
<code>IN_CLASS_ID</code>	48	80	32	yes
<code>IN_CLASS_00D</code>	48	80	32	yes
<code>OFF_GRID_00D</code> (mild)	48	80	32	yes
<code>OFF_GRID_ID</code> (severe)	0	0	0	no

against a materiality threshold of 8 M. **"640 truths" describes the bank, not uniform success across its regimes**; severe mismatch remains negative. The four fitting-family successes are a separate breakdown and are not a restatement of the regime result.

The metric is the **registered baseline-inclusive field metric** of appendix A.1, level dominated: the level fraction of the truth is 0.9838 under the level/structure projector split. It is a field metric, not a coefficient-space metric — an estimator that operates on coefficients does not thereby determine what its evaluation loss measures.

7.2 Structure is a separate, later onset

The structure-only stable span is **zero at `SNR_0 = 100` for every arm** and first becomes nonzero at `SNR_0 = 30000` , with 40 M direct and 76 M resolved. This is not "cleared at ten times the reference SNR."

Separately, R1L stage-2R validation at `SNR_0 = 1000` meets its aggregate structural-materiality conditions while its stable-span gain remains zero. It is neither a sealed main nor a stable-history-span success, and it does not replace the onset above.

7.3 Morphology: the HMT-2 sealed main

60 held-out histories — ten per family across six families — with four paired noise draws each. The draws are not 240 independent histories: the independent resampling unit is the truth. Materiality was fixed before the bank was drawn: median relative reduction ≥ 0.10 and bootstrap lower bound ≥ 0.05 . On `L896_radial_enriched` , `RESOLVED_PHYSICAL` , against the analytic source target:

estimator	gain	lower bound	material
Ridge	+0.164	+0.116	yes
TSVD	+0.133	+0.101	yes

The tested all-order flux readout does not reproduce the material improvement, and the index-sum arm reaches at most +0.015 with a negative lower bound.

Attribution, stated carefully. The labelled-stack estimates improve in this declared experiment. The tested flux readout does not reproduce that gain. The index-sum control does not establish how a physically unresolved image would perform. No causal claim about physical order resolution follows.

7.4 The qualifications that are part of the result

- `MULTI_FEATURE_RECOVERY_NEGATIVE` — reliable multi-feature recovery not achieved.
- `STABLE_MORPHOLOGY_INTERVAL_NEGATIVE` — the stable morphology interval is zero.
- `FAMILY_HETEROGENEITY` — the genuine same-class per-family result: **5 of 12 family-estimator cells** are material on the physical end-to-end target and **4 of 12** on both targets. `circular_hotspot_trajectory` is **negative under both estimators** (−0.022 ridge, −0.051 TSVD). Ten truths per family makes every interval wide, so these point estimates and intervals are reported as measured while no strong universal family claim follows in either direction; the heterogeneity is what is recorded, not a ranking.
- `CLASS_DEPENDENCE` — a separate result, not family heterogeneity. On the representation-limited control class `L448_contrast` the TSVD gain is +0.061 with lower bound +0.033 and is not material, against +0.133 / +0.101 on `L896_radial_enriched` . This changes the representation class, not the source family.
- `DIRECT_BASELINE_SATURATION_QUALIFICATION` — part of the margin comes from where the direct baseline saturates.
- Posterior calibration failed and is preserved as a failure. That failure belongs to the **R1 probabilistic estimator program** — the state-space and Wiener branch, whose joint calibration gate failed literally at 0.497 against a frozen lower bound of 0.5. It is not a posterior supplied by TSVD or ridge, which are retained as point estimators.

8. Numerical validation, and what it does not cover

component	state
Hull tessellation	9.592e-08 band-normalised geometric discrepancy between the named tessellation refinement levels, against the 1e-6 numerical-integration component budget. This is the tessellation component , not the entire hull error.
Cell measure (C13)	Repaired. Distinct from the G10q and C02 conventions; the repaired sidecars did not reweight all historical results.
Leaf assembly	Parent overlap reproduced per (detector cell, parent) to 6.9e-18.
Sign-aware envelopes	The four-endpoint product is a valid enclosure for supplied intervals; it is not the only one.
Path-domain comparator	Validated on 66 development and 192 confirmation IDs, zero forced labels. Bounded independence: primary and reference share roots, angular crossing and path classification.
Transfer quadrature	Not qualified.

Measured as whitened detector vectors on identical supported geometry, the coarse and fine representations of the transferred field differ by a **maximum relative same-support discrepancy over the 40 transferred diagnostic columns** of 9.594e-03, 4.274e-02 and 5.116e-01 at orders 0, 1 and 2, against a 5e-4 component budget. These are per-order maxima over 40 correlated columns, not one identical error on every column; separately, every one of those columns exceeds the tested threshold at every order. **This is a discrepancy between two discretisations, not a measured continuum error**, and the agreement of neither establishes the accuracy of either.

Coverage: the order-2 comparison covers 61.7% of emitting nodes and 62.8% of emitting area; 0.6092 M² carries no response bound of any kind.

Error-bound decomposition, channelwise maxima with $K = W \cdot 0 : A/E$ is 1.10, 4.57 and 4.71 and T/A is 20.02, 9.02 and 6.73. These are separate maxima, so their product is not the maximum of the product and they do not identify the dominant factor for any single channel. Cancellation is not the whole story and is not absent. Group remainder radii are *one* route to a cancellation-aware certificate, not a universal requirement.

9. Limitations and dependencies

1. **Transfer quadrature.** Section 8. Qualification needs a signed per-channel response comparison inside 5e-4, full-domain coverage or a bounded remainder, a stated accuracy for the reference, and a funded independent validation budget. Group remainder radii from outside the certified pair are **one possible** numerical certificate, consistent with section 8; any alternative construction still needs a justified reference error, controlled omitted support, and validation appropriate to itself.

2. **Order-resolution attribution.** Requires a comparison against an appropriate physical control and a new result. A quadrature pass elsewhere does not supply it.
 3. **R2 count.** Legacy measure; corrected-measure recomputation not performed.
 4. **The realized R1 metric floor η .** The rule is source-resolved and restated in appendix A.1: η is 0.05 times the median positive windowed truth norm on the prior-fit split, frozen before scoring, common to all truths. **The realized scalar has not been located in the inspected records.** `metrics.eta_value` is `null` in the executed freeze; the runner computes η from the regenerated prior-fit bank and prints it to six significant digits, and no stdout capture, run sidecar or cached prior-fit norm carrying it was found. The saved main outputs record medians of the normalised and absolute errors separately, and a quotient of two independent medians is not the per-row denominator, so η cannot be read back from them either. This limits exact reproducibility of the R1 normalisation for the claims in sections 7.1 and 7.2 and appendix A.1. It does not change the archived benchmark values, which were computed with the frozen scalar, and the floor binds only where the windowed truth norm falls below it.
 5. **Reach.** Detectability is not estimator recovery.
 6. **Enrichment.** Rank findings are model-specific, not a theorem.
 7. **Negative results retained:** severe off-grid $0 \rightarrow 0$ M, zero stable morphology interval, negative multi-feature recovery, family heterogeneity, failed posterior calibration, and the closed negative composite-representation test.
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10. Conclusion

The calculations distinguish access to earlier source times, finite-model identifiability, and recovery of a declared source object. Support-disjoint temporal functions generate exact blind directions for the sampled direct operator, while added order-labelled observations can provide nonzero historical responses. The finite-SNR dimension and estimator performance still depend on source resolution, noise, nuisance parameters and the specified loss.

The archived level and aggregate morphology improvements are bounded computational results. They neither recover a stable historical movie nor establish the unresolved-image attribution. The physical transfer-integral accuracy and an appropriate matched physical control remain separate requirements for any stronger acquisition claim.

11. References

External literature

Verified in review 039 at the level stated for each entry. Metadata and author-posted abstracts were read; full texts were not, and the narrow use listed is what each citation supports here. `BIBLIOGRAPHY_VERIFICATION_039.json` records the verification level entry by entry.

- **[GL2020]** S. E. Gralla and A. Lupsasca, *Lensing by Kerr Black Holes*, arXiv:1910.12873v2; Phys. Rev. D **101**, 044031 (2020); DOI 10.1103/PhysRevD.101.044031. Used for: the highly bent Kerr-image hierarchy has demagnification, rotation and relative-delay structure.
- **[J2020]** M. D. Johnson et al., *Universal Interferometric Signatures of a Black Hole's Photon Ring*, arXiv:1907.04329v2; Science Advances **6**, eaaz1310 (2020); DOI 10.1126/sciadv.aaz1310. Used for: photon subrings have characteristic long-baseline interferometric signatures.
- **[AART2023]** A. Cárdenas-Avendaño, A. Lupsasca and H. Zhu, *Adaptive Analytical Ray Tracing of Black Hole Photon Rings*, arXiv:2211.07469v4; Phys. Rev. D **107**, 043030 (2023); DOI 10.1103/PhysRevD.107.043030. Used for: the analytic Kerr ray-tracing framework with adaptive image sampling underlying the archived maps.
- **[H2021]** S. Hadar, M. D. Johnson, A. Lupsasca and G. N. Wong, *Photon Ring Autocorrelations*, arXiv:2010.03683v3; Phys. Rev. D **103**, 104038 (2021); DOI 10.1103/PhysRevD.103.104038. Used for: a photon-ring intensity-correlation observable for stochastic equatorial emission whose structure contains lens and source information.
- **[T2020]** P. Tiede, H.-Y. Pu, A. E. Broderick, R. Gold, M. Karami and J. A. Preciado-López, *Spacetime Tomography Using The Event Horizon Telescope*, arXiv:2002.05735v2; linked DOI 10.3847/1538-4357/ab744c. Used for: related tomography work using a constrained hotspot model in simulated recovery and spacetime inference. Journal volume and page fields were not separately verified.
- **[B2018]** K. L. Bouman, M. D. Johnson, A. V. Dalca, A. A. Chael, F. Roelofs, S. S. Doeleman and W. T. Freeman, *Reconstructing Video from Interferometric Measurements of Time-Varying Sources*, arXiv:1711.01357v2. Used for: related dynamic VLBI imaging of evolving sources under a Gaussian Markov model. Journal pagination was not verified.
- **[L2024]** A. Levis, A. A. Chael, K. L. Bouman, M. Wielgus and P. P. Srinivasan, *Orbital Polarimetric Tomography of a Flare Near the Sagittarius A^{*} Supermassive Black Hole*, arXiv:2310.07687v2. Used for: model-dependent 3-D flare-emission reconstruction combining a neural representation with a gravitational model. Journal metadata was not verified.

Scope of the literature audit. Seven primary arXiv records were verified in review 039 at metadata and abstract level. This is not a full-text audit, not a systematic novelty survey, and not a verification of the remaining entries of the archived manuscript's older reference list. No novelty or priority claim in this paper rests on that audit; the outstanding items are listed in `SUBMISSION_GAPS_039.md`.

Repository evidence

1. `artifacts/reports/E3C_GEOMETRY_WIDE_OPERATOR_AUDIT.md` — geometry-wide operator audit; reach statistics and their definitions.
2. `artifacts/reports/E3C_MECHANISM_DECOMPOSITION.md` — delay and spatial substitutions.
3. `artifacts/reports/E3D_SOURCE_CLASS_STRESS.md` — the four-class enrichment ladder.
4. `artifacts/reports/R1_HELD_OUT_MAIN.md` — sealed level main, regime table, level/structure split, structural onset.
5. `docs/Paper_I_v1_Current_Evidence_Ledger.md` — R1L stage-2R and HMT-2 sealed-main entries.
6. `artifacts/revisions/mahaka1_v4_1/R2REPLAY_20260906T144528Z_b873908/` — reference-geometry information and replay.
7. `artifacts/revisions/mahaka1_v4_1/LOC025_20260906T184213Z/LOCALIZATION_READBACK.json` — source-function localization of the two modes.

8. `artifacts/configs/R1_MAIN_FREEZE.json`, `artifacts/configs/E3C_OPERATOR_GRID_FREEZE.json` and `artifacts/configs/HMT2_SEALED_MAIN_V1.json` — the executed freezes.
9. `docs/revisions/mahakal_v4_1/PAPER_I_DEFECT_AMENDMENT_023.md` — control interpretation dispositions.
10. Rulings 026–039 under `docs/revisions/mahakal_v4_1/`.

12. Reproducibility

Every number is registered in `CLAIM_EVIDENCE_MATRIX_039.json` with its primary artifact path, commit, sha256 and row selector, and with the ten pins resolved per experiment. A hash proves which bytes were read; it does not prove the prose describes the experiment correctly, and the matrix records that distinction. `METHOD_DEPENDENCY_STATUS_039.json` carries the method-input status separately, including the one input — the realized η of section 9 — that a schema pass must not mark complete.

The editorial history of this text, including the passages withdrawn in revisions 036 through 038 and the reasons for each, is preserved in `MANUSCRIPT_CORRECTION_LEDGER_039.md` rather than in the body above.

Author review is required before any scope or submission decision.

Appendix A. Methods

Definitions recovered from the implementations that produced the results, at the commits recorded in `CLAIM_EVIDENCE_MATRIX_039.json`. Nothing here is reconstructed from general convention, and a value absent from the execution record is marked as a dependency rather than filled from a function default.

A.1 The R1 baseline-inclusive field metric

Let D be the C224 synthesis matrix on the fixed source evaluation grid and v_j the rendered truth on that same grid. For age a , the normalised Gaussian window weights are

$$w_{ap} = \exp[-(t_p + a)^2 / (2h^2)] / \exp[-(t + a)^2 / (2h^2)] \parallel_2, h = 3 M$$

and with $W_a = \text{diag}(w_{ap})$ the registered relative error is

$$E_{jb}(a) = \parallel W_a (D \hat{c}_{jb} - v_j) \parallel_2 / \max(\parallel W_a v_j \parallel_2, \eta)$$

where b labels the noisy realisation. This is a **weighted field loss on a finite evaluation grid**. The implementation computes it through the algebraically equivalent quadratic form

$$\begin{aligned} \parallel W_a (D \hat{c} - v) \parallel^2 &= \hat{c}^T M_a \hat{c} - 2 \hat{c}^T p_a(v) + s_a(v) \\ M_a &= D^T W_a^2 D, p_a(v) = D^T W_a^2 v, s_a(v) = v^T W_a^2 v \end{aligned}$$

for speed. That shortcut does not make it a coefficient-space loss.

The evaluation grid. The grid is fixed by the archived R1 caller and scorer at execution commit 5f557fb6 :
`scripts/run_r1_main.py` calls `evaluation_grid(basis.r_inner, basis.r_outer, basis.t_min, basis.t_max)` with no size overrides, and the imported `phrt.metrics.scoring.evaluation_grid` at that same commit takes `n_r = 10` , `n_phi = 12` , `n_t = 40` . The field is therefore scored at $10 \times 12 \times 40 = 4,800$ **evaluation points** with the declared equal grid weights: ten radii logarithmically spaced over `[1.8660386527060988, 49.98205255591607]` M with both endpoints included, twelve azimuths uniformly spaced over `[0, 2 $\pi$ )` with the repeated endpoint omitted, and forty source times uniformly spaced over `[-128.82234649196255, 29.0]` M with both endpoints included, flattened in `meshgrid(indexing="ij")` C order so that time varies fastest, then azimuth, then radius. This is a source-code-defined scoring grid; it is not by construction an approximation to a continuum `r dr d $\phi$  dt` norm, and no independent continuum quadrature accuracy is implied. `M_a` is the scoring matrix and must not be confused with `H_T` , the source-function Gram of section 6, or with geometric screen-cell areas.

Endpoint. On the fixed 4 M age grid (half width 3.0 M, grid maximum 120 M), the primary endpoint is the largest passing span connected to the frozen anchor, with `$\epsilon = 0.25$` and `$q = 0.95$` : the running maximum of `E_jb` over the interval must be at most `$\epsilon$` on at least `$q$` of the empirical joint truth/noise evaluations. Eight noisy draws per truth, plus one noiseless control that the endpoint excludes. Uncertainty is paired truth-cluster resampling — every draw travels with its sampled truth — 10,000 resamples, 95% intervals, frozen seed

1. The prespecified improvement floor is 8 M. `IN_CLASS_ID` is the primary regime; the sealed bank is 640 records split 256 / 64 / 256 / 64 across `IN_CLASS_ID` , `IN_CLASS_00D` , `OFF_GRID_ID` , `OFF_GRID_00D` .

Remaining dependency. `$\eta$` is a common floor fixed from the prior-fit split by the registered rule — 0.05 times the median positive windowed truth norm on that split — not chosen per truth and not taken from the test set. Its realized scalar is **not located in the inspected records**, as set out in section 9 and `METHOD_DEPENDENCY_STATUS_039.json` . The rule is resolved; the number is not.

A.2 The HMT-2 morphology metric

The metric is **state dependent**, not a single norm. Each truth-age state receives its reconciled source-side label before any arm comparison, and every state contributes to the primary endpoint:

reconciled state	error
<code>SINGLE_RESOLVED</code> , <code>MULTI_RESOLVED</code>	unbalanced assignment cost, normalised and clipped to its declared worst-case scale
<code>BLENDED</code> , <code>AMBIGUOUS</code>	normalised centroid, size, azimuthal-mode and contrast descriptor discrepancies
<code>DEAD</code>	amplitude error

`all_state_error` is a mean over state errors. The `TOTAL_FLUX` control is built with an all-ones order mixer plus a total-flux collapse, which is not the E3C per-order flux readout despite the shared word.

The paired reduction, and the order it is taken in. Write `e[j,b,a,k]` for the state-dependent error of history `j` , draw `b` , arm `a` and declared age `k` , at fixed class, estimator, SNR and target. The caller averages the state errors over the declared ages for each

draw, then averages those per-draw scores over the four noise draws of that history:

$$\bar{e}[j,a] = \text{mean}_b(\text{mean}_k e[j,b,a,k])$$

Histories are paired across arms by their fixed (family, index) identifiers, and the direct-relative reduction of each pair uses the direct history's mean error as denominator, with the recorded numerical floor:

$$r[j] = (\bar{e}[j,direct] - \bar{e}[j,a]) / \max(|\bar{e}[j,direct]|, 1e-300)$$

$$\text{reported point estimate} = \text{median}_j r[j]$$

The paired ratios are therefore formed **after** within-history averaging, and the headline is the **median across histories** — not the relative reduction of two population mean errors, and not a median of per-state ratios. With an equal number of ages and draws the two within-history arithmetic means commute; the nonlinear ratio and the outer median do not generally commute with averaging. The median's 95% interval is the 2.5th and 97.5th percentiles of 10,000 bootstrap resamples of histories at helper seed 20260954 (the supplied bank seed 20260953 plus one); the history is the independent resampling unit and its four draws are already contained in its score. The helper also computes a cell-balanced mean, which is a separate estimand and is not the declared headline. The paired wrapper intersects history identifiers and masks nonfinite paired scores; the reported main count is 60, and that count was not independently recomputed from the stored arrays here.

Scope. The archived PHYSICAL_END_TO_END label means fidelity to the analytic source. It is not validation of a real instrument and not validation of the continuous transfer quadrature.

A.3 Nuisance-adjusted information

As in section 6.1, with $H_T = R^T R$ the Gram matrix of the specified target source functions **under the declared source measure** — distinct from M_a above and from screen-cell areas. Operational directions are counted from the singular values at the declared amplitude threshold; if σ is already inside C they are not multiplied by an SNR label again.